



Undirectional and Bidirectional LSTM Models for Short-Term Traffic Prediction

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Summary

- **Traffic Flow:** Focus on short-term prediction with BiLSTM, showing superior accuracy
- **Data:** Real-world traffic data from Australian freeways used for model development
- **Architecture:** Utilized Uni-LSTM and BiLSTM with multiple layers; 4-layer BiLSTM model optimized
- **Results:** BiLSTM outperforms Uni-LSTM, deep BiLSTM models improve accuracy
- **Keywords:** Traffic Flow Prediction, BiLSTM, Uni-LSTM

Abstract

- The paper develops and evaluates short-term traffic prediction models using unidirectional (Uni-LSTM) and bidirectional (BiLSTM) deep learning long short-term memory neural networks
- BiLSTM outperforms Uni-LSTM in predicting variable horizons of speed and traffic flow, based on a common dataset from multiple freeways in Australia
- Stacked and mixed Uni-LSTM and BiLSTM models, particularly a 4-layer BiLSTM network, demonstrate improved accuracy for 15-minute prediction horizons
- The optimized 4-layer BiLSTM model achieves high prediction accuracy exceeding 90% for speeds up to 60-minute prediction horizons and above 90% accuracy for 5- and 10-minute flow prediction horizons, providing road operators with robust models tested on different freeways in Australia

Introduction

- Freeway short-term traffic prediction models are crucial for optimizing freeway operations and preventing traffic breakdowns, utilizing statistical techniques and artificial intelligence methods such as neural networks
- Recent advancements in deep learning, particularly with models like Uni-LSTM and BiLSTM, have shown promise in accurately predicting traffic dynamics, with BiLSTM offering improved accuracy by considering both past and future instances simultaneously
- This paper assesses the performance of Uni-LSTM and BiLSTM for different time horizons using speed and flow data from multiple freeways in Australia, addressing questions regarding the impact of training direction, layer addition, and model validation across different freeway datasets

Literature Review

- Short-term traffic prediction is vital for intelligent transport systems, with methodologies categorized into parametric and nonparametric approaches, where nonparametric methods like LSTM and BiLSTM show promise in handling dynamic traffic patterns

- LSTM and BiLSTM models have been extensively studied for traffic prediction, demonstrating superior performance compared to traditional methods like ARIMA and support vector regression, particularly in handling traffic flow and speed prediction under various urban driving conditions
- While LSTM models have been widely explored in the literature, few studies have addressed the potential of BiLSTM models for traffic time series prediction considering backward temporal dependencies
- This paper contributes by comparing stacked and mixed BiLSTM and LSTM layers, discussing model applicability across different locations, and utilizing field data with diverse traffic characteristics

Study Methodology

- The study utilized real-world traffic data collected from multiple freeways in Australia, including speed and flow measurements, to develop and evaluate prediction models
- Data from the Pacific Motorway, Tullamarine Freeway, and South Eastern Freeway were used, comprising various traffic conditions such as peak, nonpeak, weekday, weekend, and incident scenarios
- Each dataset was divided into training and testing subsets, with observations ranging from normal traffic conditions to those including incidents like road crashes, broken down vehicles, etc., to improve the robustness of prediction models
- The study implemented both unidirectional and bidirectional LSTM networks for short-term traffic prediction, with bidirectional LSTM offering advantages in capturing long-term dependencies by training input data in both forward and backward directions
- The LSTM networks were structured with four layers: sequence input layer, Uni-LSTM/BiLSTM layers, fully connected layer, and regression layer, with hyperparameters optimized to achieve the best accuracy
- Training and testing data were standardized to prevent overfitting, and various hyperparameter settings were tested to find the optimal combination

Comparative Evaluation of Uni-LSTM and BiLSTM

- The study evaluated the performance of Uni-LSTM and BiLSTM models for speed and flow prediction on the Tullamarine Freeway in Melbourne and the Pacific Motorway in Brisbane, using the mean absolute percentage error (MAPE) to assess accuracy
- BiLSTM outperformed Uni-LSTM in speed prediction for both freeways, achieving accuracies above 92.6% up to 60 minutes into the future for the Tullamarine Freeway
- For the Pacific Motorway, BiLSTM outperformed Uni-LSTM up to 45 minutes, with similar accuracy for the 60-minute prediction horizon

Deep and Mixed Unidirectional and Bidirectional LSTM

- The study explored the use of multiple layers of Uni-LSTM and BiLSTM, as well as combinations of both, to improve speed and flow prediction accuracy, particularly focusing on the 15-minute prediction horizon for the Tullamarine Freeway and Pacific Motorway
- Results demonstrated that deep BiLSTM models with combined layers outperformed Uni-LSTM and deep Uni-LSTM models for the 15-minute prediction horizon on both freeways, achieving accuracies of up to 99.99%, specifically, the 4-layer BiLSTM model showed the best performance compared to other configurations

Deep BiLSTM Model Validation and Transferability

- The section presents results on model validation and the potential for transferring models to other freeways without the need for recalibration and retraining, which could reduce deployment costs and resources
- Validation experiments showed promising results, with model accuracies ranging from 99.7% for 5-minute speed forecasting horizons to 71.2% for 60-minute flow forecasting horizons
- The model's performance was maintained when validated on an independent dataset from the Pacific Motorway in Brisbane, with accuracies ranging from

99.7% for 5-minute speed forecasting horizons to 82.1% for 30-minute flow forecasting horizons

Conclusions, Contributions, and Future Research Directions

- The study developed unidirectional and bidirectional LSTM networks to predict speed and flow on freeways, achieving superior performance with bidirectional LSTM models
- Results demonstrated the challenges of predicting traffic flow compared to speed, attributed to the noisy nature of flow measurements
- Extending the models with multiple LSTM and BiLSTM layers improved performance, with the 4-layered BiLSTM model outperforming others for both speed and flow prediction
- Model validation showed promising results for transferability, indicating the potential to develop transferable models with sufficient data representing diverse traffic conditions from different cities worldwide